EXPERIMENT 6b: TIME SERIES DATA ANALYSIS AND FORECAST

LEARNING OBJECTIVES:

- To understand time series data and its components.
- To analyze trends and seasonal patterns in airline passenger data.
- To perform stationarity testing using the ADF test.
- To make a non-stationary series stationary using differencing.
- To build an ARIMA forecasting model.
- To forecast future passenger numbers.
- To evaluate forecasting accuracy using MSE, MAE, and RMSE.

METHODS / PYTHON LIBRARIES USED:

Library/method	Purpose	Reason for use
Pandas	Load and manage time series data	Handles date-based data efficiently
Matplotlib	Create time series graphs	Visualizes trends and forecasts
ADF Test	Check stationarity	Determines whether differencing is required
ARIMA	Time series forecasting	Predicts future passenger values
MSE / MAE / RMSE	Forecast evaluation	Measures prediction accuracy

PROGRAM CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
from statsmodels.tsa.stattools import adfuller
from statsmodels.tsa.arima.model import ARIMA
from sklearn.metrics import mean_squared_error, mean_absolute_error

url =
"https://raw.githubusercontent.com/jbrownlee/Datasets/master/airline-passengers.csv"

df = pd.read_csv(url, parse_dates=["Month"], index_col="Month")
```

```
print(df.head())
print(df.shape)
print(df.info())

df.plot()
plt.title("Airline Passenger Data")
plt.show()

result = adfuller(df["Passengers"])
print("ADF p-value:", result[1])

df["Diff"] = df["Passengers"].diff()
diff = df["Diff"].dropna()

result = adfuller(diff)
print("Differenced ADF p-value:", result[1])

diff.plot()
plt.title("Differenced Series")
plt.show()

train = df["Passengers"][:-12]
test = df["Passengers"][-12:]

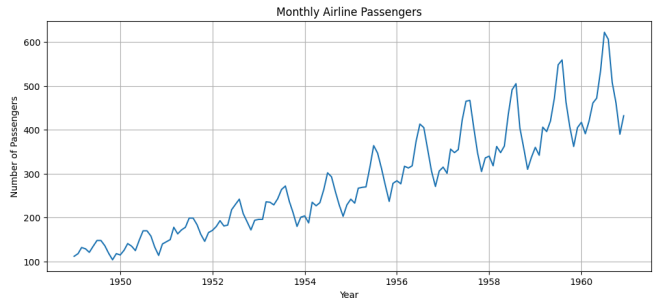
print("RMSE:", rmse)

plt.plot(train, label="Train")
plt.plot(test, label="Test")
plt.plot(test.index, pred, label="Prediction")
plt.legend()
plt.title("ARIMA Forecast")
plt.show()
```

OUTPUT:

```
plt.xlabel("Year")
plt.ylabel("Number of Passengers")
plt.legend()
plt.grid()
plt.show()
```

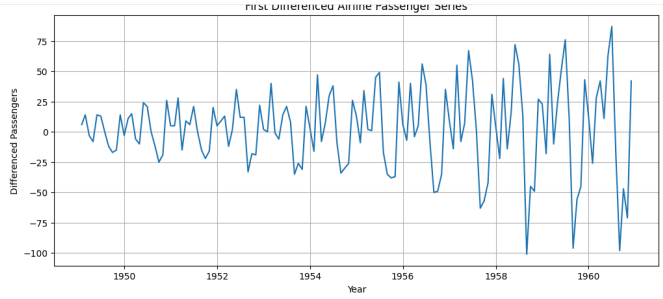
```
Missing Values:
#Passengers    0
dtype: int64
```



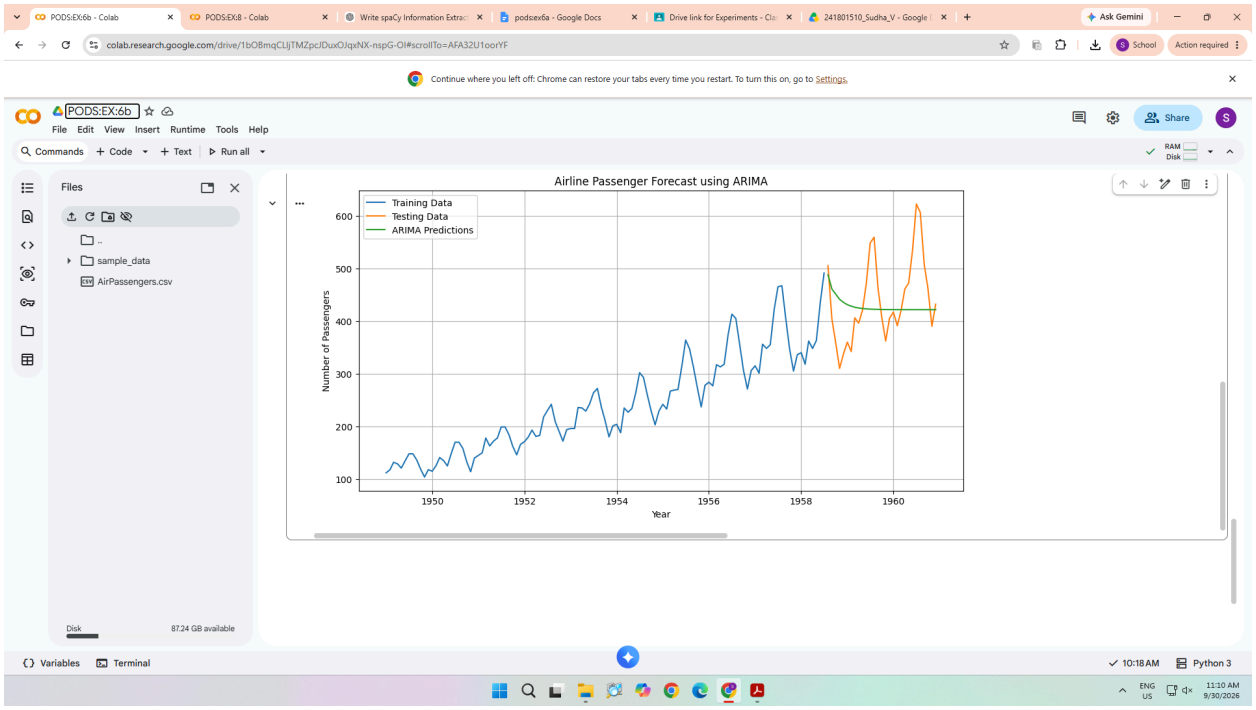

```
/tmp/ipykernel_1175/2629396077.py:37: FutureWarning: adfuller currently returns a plain tuple whose length depends on the store and autolag arguments. In release 0.16 or after July
  result = adfuller(df["#Passengers"])

ADF Test - Original Series
ADF Statistic: 0.8153688792060498
```


```
plt.xlabel("Year")
plt.ylabel("Number of Passengers")
plt.legend()
plt.grid()
plt.show()
```


```
/tmp/ipykernel_1175/2629396077.py:63: FutureWarning: adfuller currently returns a plain tuple whose length depends on the store and autolag arguments. In release 0.16 or after July
  result_diff = adfuller(df_diff)
/usr/local/lib/python3.13/dist-packages/statsmodels/tsa/base/tsa_model.py:480: ValueWarning: No frequency information was provided, so inferred frequency MS will be used.
  self._init_dates(dates, freq)
/usr/local/lib/python3.13/dist-packages/statsmodels/tsa/base/tsa_model.py:480: ValueWarning: No frequency information was provided, so inferred frequency MS will be used.
  self._init_dates(dates, freq)
/usr/local/lib/python3.13/dist-packages/statsmodels/tsa/base/tsa_model.py:480: ValueWarning: No frequency information was provided, so inferred frequency MS will be used.
  self._init_dates(dates, freq)
```



LEARNING OUTCOME:

- Analyze monthly time series data using Python.
- Identify trends and seasonal patterns.
- Perform the Augmented Dickey-Fuller test for stationarity.
- Apply differencing to non-stationary data.
- Build an ARIMA forecasting model.
- Predict future airline passenger demand.
- Evaluate forecasts using MSE, MAE, and RMSE.
- Visualize actual and predicted passenger values.